The documentation and process conversion measures necessary to comply with this document shall be completed by 12 June 2024.

INCH-POUND

MIL-PRF-19500/528C
w/AMENDMENT 1
12 March 2024
SUPERSEDING
MIL-PRF-19500/528C
5 December 2018

PERFORMANCE SPECIFICATION SHEET

* TRANSISTOR, NPN, SILICON, POWER,
TYPES 2N6032 AND 2N6033,
JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, power transistors. Three levels of product assurance (JAN, JANTX, and JANTXV) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outlines. The device package outlines are as follows: TO-3 in accordance with figure 1 for all encapsulated device types.

1.3 Maximum ratings.

Types	P_T (1) T_C = +25°C	$R_{\theta JC}$	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}
	W	°C/W	V dc	V dc	V dc	A dc	A dc	°C
2N6032	140	1.25	120	90	7.0	10	50	-65 to +200
2N6033	140	1.25	150	120	7.0	10	40	-65 to +200

(1) Between T_C = +25°C and T_C = +200°C, linear derating factor (average) = 800 mW/°C.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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FSC 5961

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1.4 Primary electrical characteristics at T_A = +25°C.

Types	h_{FE1}				C_{obo}		$\|h_{fe}\|$	
	I_C = 50 A dc V_{CE} = 2.6 V dc		I_C = 40 A dc V_{CE} = 2.0 V dc		.1 MHz ≤ f ≤ 1 MHz I_E = 0 A dc V_{CB} = 10 V dc		f = 5 MHz I_C = 2.0 A dc V_{CE} = 10 V dc	
					pF			
	Min	Max	Min	Max	Min	Max	Min	Max
2N6032 2N6033	10	50	10	50		1,000 1,000	10 10	40 40

Type	$V_{BE(sat)}$				$V_{CE(sat)}$				switching			
	I_C = 50 A dc I_B = 5 A dc		I_C = 40 A dc I_B = 4 A dc		I_C = 50 A dc I_B = 5 A dc		I_C = 40 A dc I_B = 4 A dc		t_{on}		t_{off}	
	V dc		V dc		V dc		V dc		(see table I and figure 4) µs			
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
2N6032 2N6033		2.0		2.0		1.3		1.0		0.5 0.5		2.0 2.0

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX” and "JANTXV".

1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

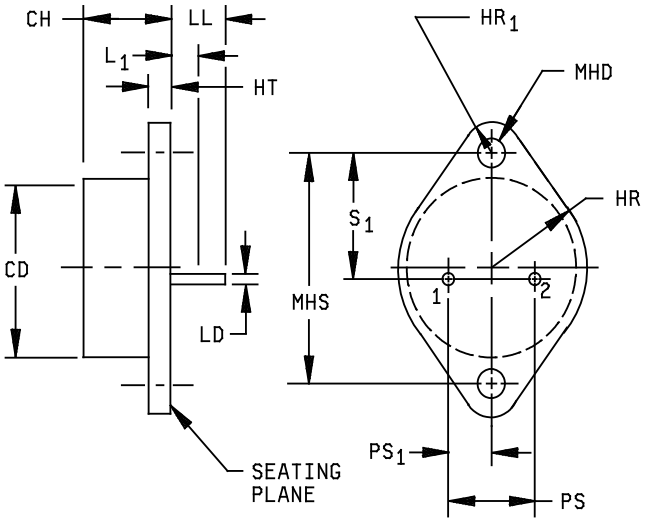
1.5.2.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "6032" and "6033".

1.5.2.3 Suffix letters. No suffix letters are used on devices that are packaged in the TO-3 package of figure 1.

1.5.3 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.875		22.22	3
CH	.250	.450	6.35	11.43	
HR	.495	.525	12.57	13.34	
HR_1	.131	.188	3.33	4.78	
HT	.050	.135	1.27	3.43	
L_1		.050		1.27	5, 9
LD	.059	.061	1.50	1.55	5, 9
LL	.312		7.92		5
MHD	.151	.161	3.84	4.09	7
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	4
PS_1	.205	.225	5.21	5.72	4, 5
S_1	.655	.675	16.64	17.14	4



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Body contour is optional within zone defined by CD.
4. These dimensions shall be measured at points .050 (1.27 mm) to .055 (1.40 mm) below seating plane. When gauge is not used, measurement shall be made at seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. Terminal 1 is the emitter, terminal 2 is base. The collector shall be electrically connected to the case.
9. LD applies between L_1 and LL. Lead diameter shall not exceed twice LD within L_1.

FIGURE 1. Physical dimensions (similar to TO-3).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1 herein.

3.4.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (table E-IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

* Screen	Measurements for JANTX and JANTXV levels
(1) 3c	Thermal impedance, see 4.3.2
9	Not applicable
11	I_{CEX1} , h_{FE1}
12	See 4.3.1
13	Subgroup 2 of table I herein; ΔI_{CEX1} = ±100% of initial value or 20 μA dc, whichever is greater; Δh_{FE1} = ±25%

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_J = +187.5°C ±12.5°C, $T_A \leq$ +35°C.

JANTX and JANTXV levels
2N6032 ----------V_{CB} = 60 V dc.
2N6033 ----------V_{CB} = 100 V dc.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW}, (and V_H where appropriate). See table II, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and 4.4.2.1 herein.

4.4.2.1 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1037	For solder die attach: $V_{CB} \geq 10$ V dc, For eutectic die attach: $V_{CB} \geq 10$ V dc, adjust P_T to achieve $T_J = +175°C$ min.
B5	3131	Thermal impedance, see 1.3.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows.

Subgroup	Method	Condition
C2	2036	Test condition A, weight = 10 pounds, t = 15 s.
C5	3131	Thermal impedance, see 1.3, (not required if performed in group B).
C6	1037	For solder die attach: $V_{CB} \geq 10$ V dc, For eutectic die attach: $V_{CB} \geq 10$ V dc, adjust P_T to achieve $T_J = +175°C$ min.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.3 Coil selection for safe operating area tests. In selecting coils for use in clamped and unclamped inductive SOAR tests, prime consideration should be given to the recommended commercially available coil. However, due to the extreme critical nature of the coil in these circuits and wide tolerance of some commercially available coils (+100, -50 percent), it shall be the semiconductor manufacturer's responsibility, to prove upon request, compliance or equivalency of any coil used (commercial or inplant designed) to be within (+20, -10 percent) of the specified inductance at the rated current and dc resistance.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to emitter		I_C = 200 mA dc; f = 30 - 60 Hz; L = 15 mH (see figure 2)	$V_{(BR)CEO}$			
2N6032				90		V dc
2N6033				120		V dc
Breakdown voltage, collector to emitter		I_C = 200 mA dc; f = 30 - 60 Hz; L = 15 mH (see figure 2)	$V_{(BR)CER}$			
2N6032				110		V dc
2N6033				140		V dc
Breakdown voltage, collector to emitter		I_C = 200 mA dc; f = 30 - 60 Hz; L = 2 mH (see figure 2)	$V_{(BR)CEX}$			
2N6032				120		V dc
2N6033				150		V dc
Emitter to base cutoff current	3061	Bias condition D V_{EB} = 7 V dc	I_{EBO}		10	mA dc
Collector to emitter cutoff Current	3041	Bias condition D V_{CE} = 80 V dc	I_{CEO}		10	mA dc
Collector to emitter cutoff Current	3041	Bias condition A; V_{BE} = -1.5 V dc	I_{CEX1}		250	µA dc
2N6032		V_{CE} = 110 V dc				
2N6033		V_{CE} = 135 V dc				
Collector to base cutoff Current	3036	Bias condition D;	I_{CBO}			
2N6032		V_{CB} = 120 V dc			25	mA dc
2N6033		V_{CB} = 150 V dc			25	mA dc
Forward-current transfer ratio	3076	Pulsed (see 4.5.1)	h_{FE1}			
2N6032		V_{CE} = 2.6 V dc; I_C = 50 A dc		10	50	
2N6033		V_{CE} = 2.0 V dc; I_C = 40 A dc		10	50	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector-emitter saturation voltage 2N6032	3071	I_C = 50 A dc; I_B = 5.0 A dc; pulsed (see 4.5.1)	$V_{CE(sat)}$		1.3	V dc
Collector-emitter saturation voltage 2N6033	3071	I_C = 40 A dc; I_B = 4.0 A dc pulsed (see 4.5.1)	$V_{CE(sat)}$		1.0	V dc
Base-emitter saturation voltage 2N6032	3306	Test condition A; I_B = 5.0 A dc; I_C = 50 A dc; pulsed (see 4.5.1)	$V_{BE(sat)}$		2.0	V dc
Base-emitter saturation voltage 2N6033	3306	Test condition A; I_B = 4.0 A dc; I_C = 40 A dc; pulsed (see 4.5.1)	$V_{BE(sat)}$		2.0	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition D;V_{BE} = -1.5 V dc; V_{CE} = 100 V dc	I_{CEX2}			
2N6032 2N6033					15 10	mA dc mA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	Pulsed (see 4.5.1)	h_{FE2}			
2N6032 2N6033		V_{CE} = 2.6 V dc; I_C = 50 A dc V_{CE} = 2.0 V dc; I_C = 40 A dc		5 5		
Subgroup 4						
Magnitude of small-signal short- circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 2.0 A dc; f = 5.0 MHz	$\|h_{fe}\|$	10	40	
Open circuit Output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		1,000	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Pulse response:	3251	Test condition A except test circuit and pulse requirements in accordance with figure 3.				
Turn-on time			t_{on}		0.5	µs
2N6032		V_{CC} = 30 V dc ±2; I_C = 50 A dc; I_{B1} = 5.0 A dc				
2N6033		V_{CC} = 30 V dc ±2; I_C = 40 A dc; I_{B1} = 4.0 A dc				
Turn-off time			t_{off}		2.0	µs
2N6032		V_{CC} = 30 V dc ±2; I_C = 50 A dc; I_{B2} = 5.0 A dc, I_{B2} = -5.0 A dc				
2N6033		V_{CC} = 30 V dc ±2; I_C = 40 A dc; I_{B1} = 4.0 A dc, I_{B2} = -4.0 A dc				
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; t = 1 s; 1 cycle; (see figure 4)				
Test 1 2N6032 only		I_C = 50 A dc; V_{CE} = 2.8 V dc				
Test 2 2N6033 only		I_C = 40 A dc; V_{CE} = 3.5 V dc				
Test 3		I_C = 5.8 A dc; V_{CE} = 24 V dc				
Test 4		I_C = 0.9 A dc; V_{CE} = 40 V dc				
Test 5 2N6032 only		I_C = 0.18 A dc; V_{CE} = 90 V dc				
Test 6 2N6033 only		I_C = 0.1 A dc; V_{CE} = 120 V dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Safe operating area (switching)	3053	Load condition C; (unclamped inductive load) (see figure 5); T_C = +25°C; Duty cycle ≤ 10 percent; R_S = 0.1 ohm;				
Test 1		$t_p \approx 10$ ms; $R_{BB1} = 1\Omega$; V_{BB1} = 10 V dc max; $R_{BB2} = 20\Omega$; V_{BB2} = 4 V dc;				
2N6032 2N6033		I_C = 50 A dc; I_C = 40 A dc; V_{CC} = 10 V dc L = 50 µH, 0.1 Ω;				
Test 2		$t_p \approx 10$ ms; $R_{BB1} = 10\Omega$; V_{BB1} = 10 V dc max; $R_{BB2} = 20\Omega$; V_{BB2} = 4 V dc; I_C = 10 A dc; V_{CC} = 10 V dc, L = 500 µH, 0.1 Ω				
Electrical measurements		See table I, subgroup 2				
Safe operating area (switching)	3053	T_A = +25°C; duty cycle ≤ 10 percent. $t_p \approx 10$ ms (vary to obtain I_C); $R_S \leq 1\ \Omega$; $V_{CC} \geq 50$ V dc; L = 50 µH, 0.1 Ω				
2N6032 2N6033		I_C = 50 A dc; I_C = 40 A dc;				

2N6032
2N6033
		clamp voltage = 90 V dc;
		clamp voltage = 120 V dc;
		(see figure 6)				

Electrical measurements		See table I, subgroup 2				
Subgroups 6 and 7						
Not applicable						

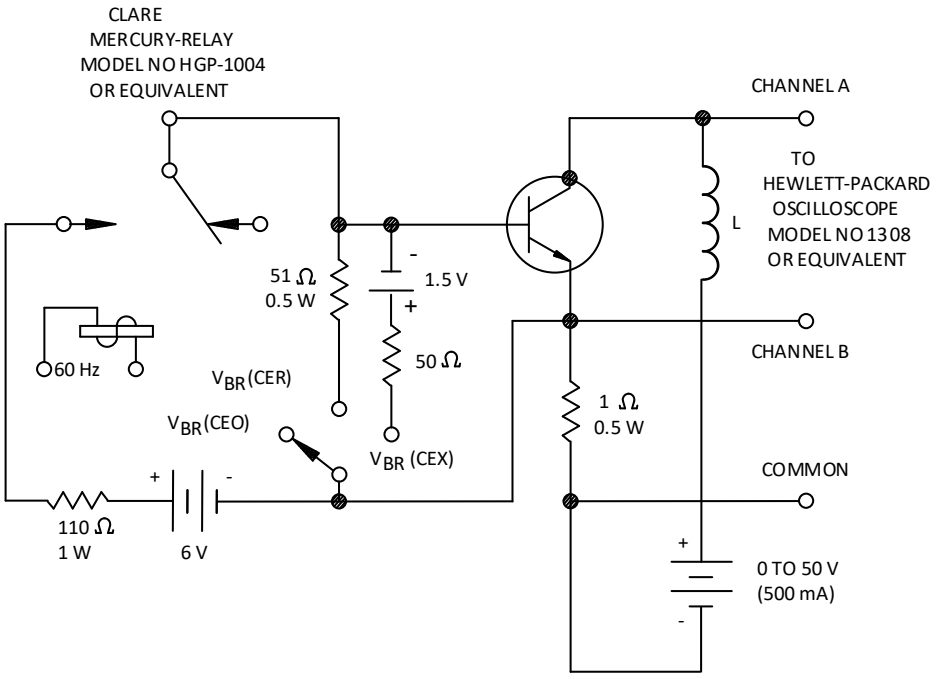
1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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TABLE II. Group E inspection (all quality levels) for qualification and re-qualification only.

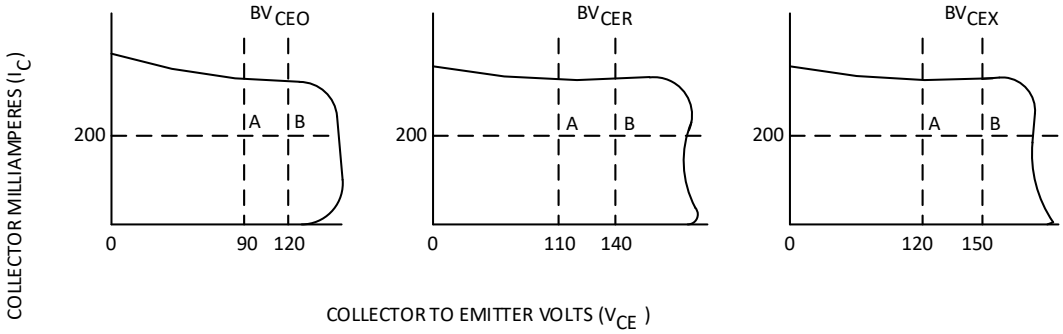
Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	500 cycles minimum	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2 herein.\	
Subgroup 2			45 devices c = 0
Blocking life	1048	Test temperature = +125°C; V_{CB} = 80 percent of rated, without going over the max rate for V_{CEO}; T = 1,000 hours.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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L=15 mH FOR $V_{BR(CEO)}$, $V_{BR(CER)}$ MEASUREMENTS
L=2 mH FOR $V_{BR(CEX)}$ MEASUREMENTS

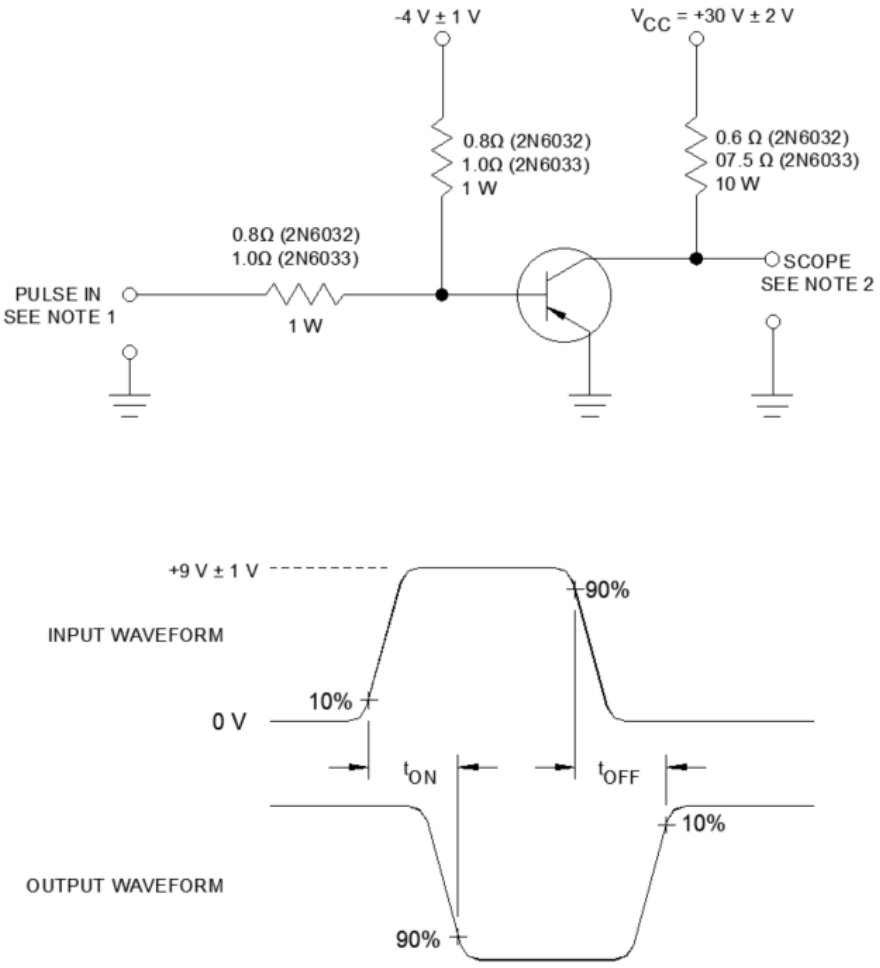
OSCILLOSCOPE DISPLAY



NOTE:
$V_{(BR)CEO}$, $V_{(BR)CER}$, $V_{(BR)CEX}$ is acceptable when the trace falls to the right and above point "A" for type 2N6032. The trace shall fall to the right and above point "B" for type 2N6033.

FIGURE 2. $V_{(BR)CEO}$, $V_{(BR)CER}$, $V_{(BR)CEX}$ measurement circuit.

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NOTES:
1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each < 20 nanoseconds; duty cycle < 2%, generator source impedance shall be 50 Ω; pulse width : 20 ns.
2. Output sampling oscilloscope: Z_{in} > 100 kΩ; C_{in} < 50 pF; rise time < 20 nanoseconds.

FIGURE 3. Pulse response test circuit.

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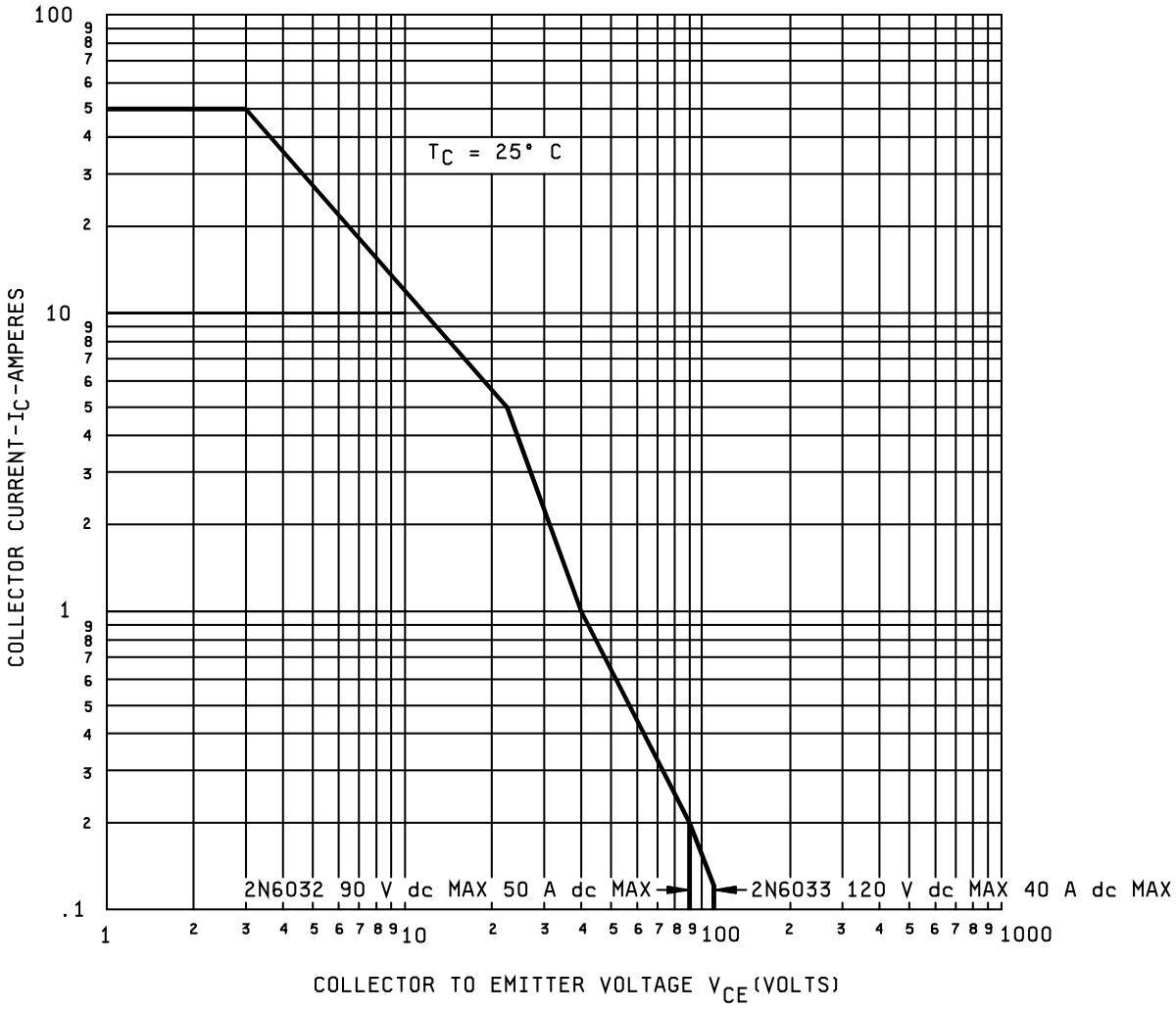


FIGURE 4. Maximum safe operating area graph (continuous dc).

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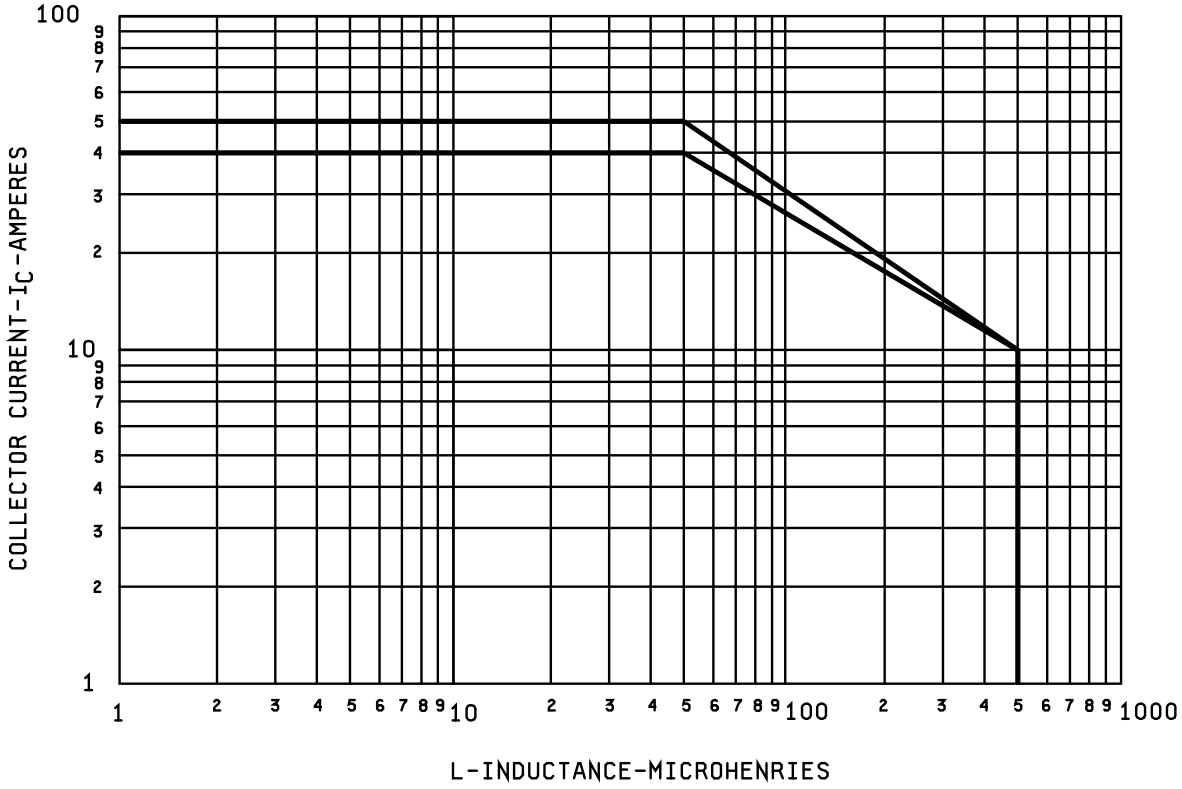
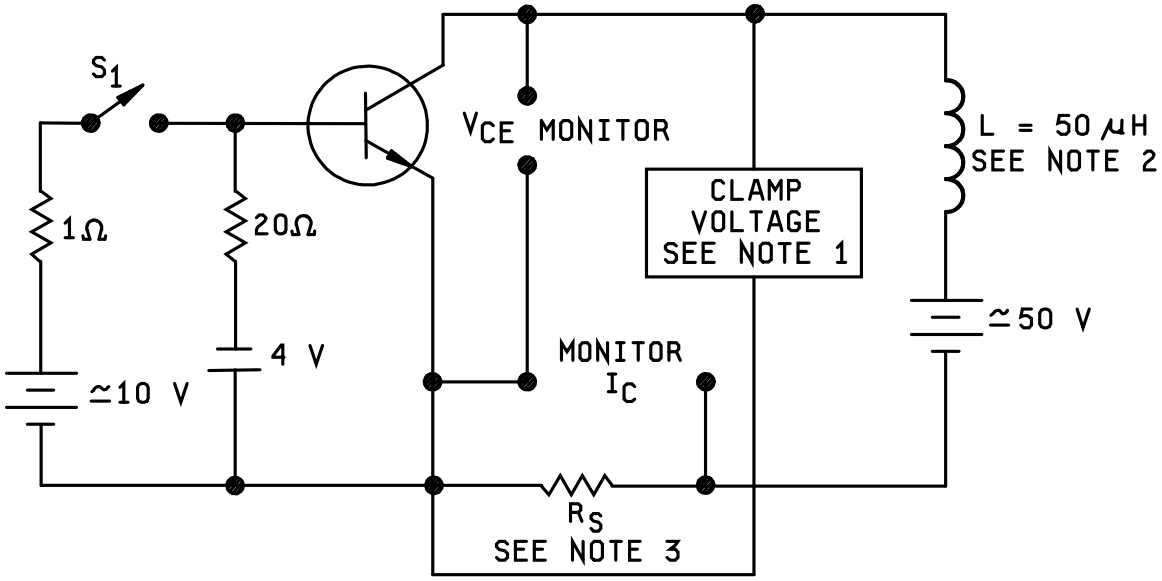


FIGURE 5. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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NOTES:
1. Either a clamping circuit or clamping diode may be used.
2. The coil used shall provide a minimum inductance of 50 μH with a maximum dc resistance of .1 ohm.
3. $R_S \leq .1\ \Omega$, 12 W, 1percent tolerance maximum (non-inductive).

Procedure

1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if clamp voltage not reached and maintained until the current returns to zero.
3. Perform specified endpoint tests.

FIGURE 6. Clamped inductive sweep test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

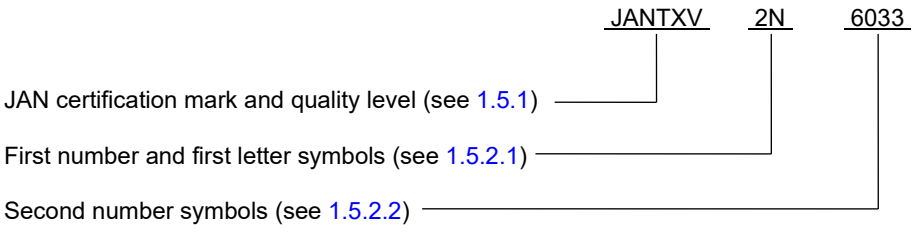
c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QPDSIS-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://qpldocs.dla.mil.

6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

PINs for type 2N6032	PINs for types 2N6033
JAN2N6032	JAN2N6033
JANTX2N6032	JANTX2N6033
JANTXV2N6032	JANTXV2N6033

6.6 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

* 6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Air Force - 85
* NASA – NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2024-026)

Review activities:
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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